

There are two trees.

This is how the first tree I found starts.

$3 \rightarrow 5 \rightarrow L=4$.

3 increments by 8

5 increments by 12.

$(2^k)-1$ increments by $2^{(k+1)}$.

$3(2^k-1)$ increments by 2^k .

This pattern will catch every positive integer 3 modulus 8, and 5 modulus 12. The pattern is always 3+8x, even, 5+12x, even, and then even, even, 4+9x

Note that

For all x.

This is the first rule for this set of even mersennes. Note that k must = 0mod2.

Now this actually has a higher and more complex structure to it, and this principle is illustrated first with 15 and 23.

In this case k = 4.

15 increments by 32.

23 increments by 16.

So, same deal for the increment rate.

This pattern will catch every positive integer 15 modulus 32, every third integer 23 modulus 16, and, notably, will always be chained with 3 modulus ... something.

Let's try to write it out.

(15, 23, 35, 53, 40). (47, 71, 107, 161, 121). (79, 119, 179, 269, 202).

We can write this as 15+32(x), even, 23+48(x), even, 3 + 32 + 72(x), even, 5 + 48 + 108(x), even, even, 4+36+81(x).

The first recurrence in this pattern always has an even L. the second always has an odd L. and then they flip back and forth.

So, before we continue, we'll always have $15 \bmod 32 \rightarrow 23 \bmod 48 \rightarrow 35 \bmod 72 \rightarrow 53 \bmod 108 \rightarrow L$ (even or odd), $40 \bmod 81$.

Ok. Let's do the next step.

The next pair is 63 and 95.

63 increments by 128.

95 increments by 64.

We can write this one as $63+128(x)$, even, $95+192(x)$, even, $15+128+288(x)$, even, $23+192+432(x)$, even, $3+32+288+648(x)$, even, $5+48+432+972(x)$, even, even, $4+36+324+729(x)$.

For $x = 0$

(63, 95, 143, 215, 285, 437, 364)

(191, 287, 431, 647, 971, 1457, 1093).

(319, 479, 719, 1079, 1619, 2429, 1822).

Ok.

So without trying anything else

I'm gonna write out what the next length of chain is

$255+512(x)$, even, $383+768(x)$, even, $63+512+1152(x)$, even, $95+768+1728(x)$, even, $15+128+1152+2592(x)$, even, $23+192+1728+3888(x)$, even, $3+32+288+2592+5832(x)$, $5+48+432+3888+8748(x)$, even, even, $4+36+324+2916+6561(x)$.

NOTE THAT

With L , the chain is actually $(4 \cdot 9^0) + (4 \cdot 9^1) + (4 \cdot 9^2) + (4 \cdot 9^3) + (4 \cdot 9^4) \dots$ so the strand always falls on this number, and then is divided by 4 to produce L .

Another way to write this would be

$255+512(x)$, even, $383+768(x)$, even, $63+512+1152(x)$, even, $95+768+1728(x)$, even, $15+128+1152+2592(x)$, even, $23+192+1728+3888(x)$, even, $3+32+288+2592+5832(x)$, $5+48+432+3888+8748(x)$, $(4 \cdot 9^0) + (4 \cdot 9^1) + (4 \cdot 9^2) + (4 \cdot 9^3) + (4 \cdot 9^4)(x)$, even, $(4 \cdot 9^0) + (4 \cdot 9^1) + (4 \cdot 9^2) + (4 \cdot 9^3) + (9^4)(x)$.

See what I mean?

I can articulate the specific rules of expansion and turn this into a simple algebraic rule I think.

Next question.

Is 7 indicative of: $3 \bmod 4$, or $7 \bmod 8$?

Likewise is 11 from $5 \bmod 6$, or from $11 \bmod 12$?

Is it possible that this is actually just powers of 4, and $k = 0 \bmod 2$ is universal?

Why don't we consider it a feature of $3 \bmod 4$.

I think the reason why we wouldn't is that the patterning is different.

OK.

We'll do $7 \bmod 8$, with the understanding that the numbers are just offset from the smaller modulus if that makes sense.

The pattern I see for this one is

$7+8x$, even, $11+12x$, even, $17+18x$, even, even, 13

This isn't mutually exclusive with the other tree.

Note how the next instance is 15, even, 23, even, 35, even, [53], ...
Which is from the other tree.

Let's try another.

22, even, 35, even, 53, even, even, 40.

Hmm.

I think these odd mersennes are attracted into the same sink of the evens.

29 is the second of (19, 29)... to 22.

Also violates the $11+12x$ rule, so that doesn't work.

They seem like they might not ever be too far from this chain, but I'm not seeing any clear order in what's presented.

Interesting.

Good to rule out.

I think I was looking at $3 \bmod 4$ last time.

So that would be $3+4x$, $5+6x$, and then L is $4+9x$.

(3, 5, 4)

(7, 11, 17, 13)

(11, 17, 13)

(15, 23, 35, 53, 40)

(19, 29, 22)

I think it'd be interesting to see if I can find some sort of structural regularity with the odd mersennes.

I still think what I have is capable of maybe catching all the OE OE... chains without me extrapolating it.

I need a counterexample.

Claude says this is the right way to state this so I'm gonna attempt to write this out

The chain is built in twos

So we can call this chain depth $n = 1$

$3+8x$, $(3+8x)/2$, $5+12x$, $(5+12x)/2$, $4+36x$, $4+18x$, $4+9x$ where $x =$ all real numbers.

I'm just gonna make the divisibility stuff E for now.

The next step of the chain is

$15+32(x)$, even, $23+48(x)$, even, $3 + 32 + 72(x)$, even, $5 + 48 + 108(x)$, even, even, $4+36+81(x)$.

It's that k is incremented by two so the shift is gonna move additively.

What the fuck is this math.

The frustrating thing is that I know exactly how it works but I have to figure out how to state it correctly

An inductive definition that says "given chain depth n , the depth $n+1$ chain is [this transformation applied to the depth n chain]"

So the inductive step you're trying to write is something like: "To build depth $n+1$ from depth n , prepend the new Mersenne pair with increments $(2^{(2n+3)}, 3 \cdot 2^{(2n+2)})$, then for each of the $2n$ existing entries, add the corresponding new-depth increment to its base and multiply its modulus by $3/2$ twice" — though you'll want to be precise about which increment gets added where.

The geometric series thing is just this: if you're adding up $1 + r + r^2 + r^3 + \dots + r^{(n-1)}$, the sum is $(r^n - 1)/(r - 1)$. So your L sequence is $4 \cdot (1 + 9 + 9^2 + \dots + 9^{(n-1)}) = 4 \cdot (9^n - 1)/(9 - 1) = 4 \cdot (9^n - 1)/8 = (9^n - 1)/2$. That's the whole trick.

So the chain depth $n = 1$ shakes out to this, right

$3+8x$, $(3+8x)/2$, $5+12x$, $(5+12x)/2$, $4+36x$, $4+18x$, $4+9x$ where $x =$ all real numbers.

Fuck the even numbers for right now, noting those as E.

Let's abstract this to the specific exponent number

I'm gonna make the exponent variable c .

So at chain depth $n = 1$, c must = 2.

$((2^c)-1)+(2^{c+1})x$, E, $(3(2^{c-1})-1)+3x(2^c)$, $(9^0)(2^c)+9(2^c)x$, $(9^0)(2^c)+9(2^{c-1})x$,
 $(9^0)(2^c)+9(2^{c-2})x$

At chain depth $n = 2$, $c = 4$.

$15+32(x)$, even, $23+48(x)$, even, $3 + 32 + 72(x)$, even, $5 + 48 + 108(x)$, even, even, $4+36+81(x)$.

$((2^c)-1)+(2^{c+1})x$, E, $(3(2^{c-1})-1)+3x(2^c)$, E, $((2^{c-2})-1)+(2^{c+1})+9x(2^{c-1})$, E,
 $3(2^{c-2})-1$

x is equivalent to all $\text{int} \geq 0$.

$c=2n$.

Chain length $n=1$, $c=2$

$2^c - 1 + 2^{c+1}x$, $(2^c - 1 + 2^{c+1}x)/2$, $3(2^{c-1}) - 1 + 3x2^c$, $9^0 2^2 + 9^1 2^2 x$, $9^0 2^2 + 9^1 2^1 x$, $9^0 2^2 + 9x$

Chain length $n=2$, $c=4$

$2^c - 1 + 2^{c+1}x$, $((2^c - 1) + 2^{c+1}x)/2$, $3(2^{c-1}) - 1 + 3x2^c$, $((3(2^{c-1}) - 1) + 3x2^c)/2$, $2^{c-2} - 1 + 2^{c+1} + 9x2^c$
 $3(2^{c-3}) - 1 + 3(2^c) + 27x2^{c-2}$, $9^0 2^2 + 9^1 2^2 + 9^2 2^2 x$, $9^0 2^2 + 9^1 2^2 + 9^2 2^1 x$, $9^0 2^2 + 9^1 2^2 + 9^2 x$

Chain length $n=3$, $c=6$

$2^c - 1 + 2^{c+1}x$, $((2^c - 1) + 2^{c+1}x)/2$, $3(2^{c-1}) - 1 + 3x2^c$, $((3(2^{c-1}) - 1) + 3x2^c)/2$, $2^{c-2} - 1 + 2^{c+1} + 9x2^c$
 $3(2^{c-3}) - 1 + 3(2^c) + 27x2^{c-2}$, $(3(2^{c-3}) - 1 + 3(2^c) + 27x2^{c-2})/2$, $2^{c-4} - 1 + 2^{c-1} + 9(2^{c-1}) + 81x2^{c-3}$, 3
 $+ 243x(2^{c-4})$, $9^0 2^2 + 9^1 2^2 + 9^2 2^2 + 9^3 2^2 x$, $9^0 2^2 + 9^1 2^2 + 9^2 2^2 + 9^3 2^1 x$, $9^0 2^2 + 9^1 2^2 + 9^2 2^2 + 9^3 x$

Note that the recurrence relationships tend to only apply to the even mersennes in this exact shape, BUT

$7 \bmod 16$ isn't mutually exclusive with the tree and its L begins at 13 and it increments by 27

So we got

$1 \bmod 4 \rightarrow$ second member is $5 \bmod 4$; meets with L at $3 \bmod 8$, 4, 4 increments by 9x

$3 \bmod 8$

$7 \bmod 16 \rightarrow$ second member is $23 \bmod 16$.

$15 \bmod 32$ L begins at 40 and increments by 81

$31 \bmod 64$ L begins at

$1 \bmod 4$ L begins at 1 increments by 3

$3 \bmod 8$ L begins at 4 increments by 9

$7 \bmod 16$ L begins at 13 increments by 27

$15 \bmod 32$ L begins at 40 increments by 81

$31 \bmod 64$ L begins at 121 increments by 243.

$63 \bmod 128$ L begins at 364 increments by 729

$1 \bmod 4$'s first L is 1, and it increments by 3, and its second increment is $5 \bmod 4$, which is when it first meets with $3 \bmod 8$

First L for 7 is 13; so $4+9$, and that's also the second time $1 \bmod 4$ and $3 \bmod 8$ align, because it's the first 3-length k-segment

All 1-length k-chains are $1 \bmod 4$.

$15 \bmod 32$

Fuck

It's indexed so that a length of consecutive compression segments of any length k is going to hit exactly that

1,E,E,1

5,E,E,4

9,E,E,7

13,E,E,10

17,E,E,13

21,E,E,16
25,E,E,19

The equation can again be restated to decrement through all the mersenne index.
Length of consecutive 1-length expansions is based on mersenne index at start paired with increment rate.

27 (famously weird small number)

First chain. Segment length of 2, so $x=3$, L is $4+9x = 31$, yes.

Compression after L was length 1; the L produced was 31 which created this chain, thus the expansion spike.

Next chain is 31 with a segment length of 5. $x=1$, L is $121+243x$.

The L is 121.

121 is a 1-length, so $L =$

Note that the compressions after an L are variable; here it was length 2, but it changes.

Next chain starts with 91. Length of 2, so L is $4+9x$, $x=11$, yes.

The L produced was 103, which is the start of another segment of length 3, meaning it's $7+16x$.
So L is $13+27x$, x is 6, yes.

Notably this chain is $7+16x$, then $3+8+24x$, then $5+12+36x$ for the three terms.

Or $1 + 16 + 36x$, to restate.

The L produced is 175.

Then this produces a 4-length k-segment chain,
So it's gonna be L is $40+81x$, L in this case is 445, so $x=5$, yes.

Note that x is mostly odd, save for 6, but that also flipped it right into another compression state

The L produced is 445, which is a 1-length, so it's $1+3x$, $x=148$

The L produced is 334, which decrements again to 167
167 so $7+16x$, x is 10, L is $13+270$, yep

L produced is 283

2-length to 319, so $3+8x$, L is $4+9x$, yep

L produced is 319, which is notably a member of $63 \bmod 128$, L is $364+729x$, $x=2$,
L's gonna be 1822, correct.

This is a predictable one for me.

Then it goes down to 911 and into a 4-length, so L is gonna be $40+81x$, x is 28, L's gonna be 2308.

Bumps down to 577, which is a 1-length, so x is 144, and L is gonna be $1+3x$, so it's gonna be 433, which it is.

433 is another 1-length, so x is gonna be 108, and L is gonna be $1+3x$, so it makes 325.

325 is also $1 \bmod 4$, with an X of 81, so it's gonna produce an L of $1+3(81)$, which is 244, which is further divisible.

Goes down to 61 at this point, another 1-length, so it's gonna make $1+3(15)$, which is 46, which is further divisible.

Lands on 23, which is the second iteration of $7 \bmod 16$, the L is $13+27(1)$ so 40, which it is.

Then 40 is further divisible to 5, which is the second iteration of $1 \bmod 4$, so the L is $1+3(1)$, which is 4, and we're done.

To summarize

All expansion operations

Length 2, fourth recurrence, produced odd L

Length 1, thirty-first recurrence, produced odd L

Length 5, first recurrence, produced odd L

Length 2, twelfth recurrence, produced odd L

Length 3, seventh recurrence, produced odd L

Length 4, sixth recurrence, produced odd L (445)

Length 1, one hundred and twelfth recurrence, produced even L (334) as YOU WOULD EXPECT.

One compression operation followed.

That's $1+3(3*37)$

Length 3, eleventh recurrence, produced odd L

Length 3, seventh recurrence, produced odd L

Length 2, thirty-sixth recurrence, produced odd L

Length 6, third recurrence, produced even L (1822) as you would expect

One compression operation followed.

Note that's gonna be $63+128(2)$.

Length 4, twenty-ninth recurrence, produced even L (2308)

This time two compression operations followed.

Length 1, one hundred and forty-fifth recurrence, produced odd L

Length 1, one hundred and ninth recurrence, produced odd L

Length 1, eighty-second recurrence, produced even L as you'd expect

[For these, note it's $(1+3(9*16))$, $(1+3(9*12))$, $(1+3(9*9))$ for x]

Two compression operations followed.

Length 1, sixteenth recurrence, produced even L

That's $1+3(3*5)$

One compression operation followed.

Length 3, second recurrence, produced even L, 40.

That's gonna be $7+16(1)$

Three compression operations followed.

Length 1, second recurrence, produced even L, 4.

Two compression operations followed.

That's gonna be $1+3(1)$.